

CLAS Collaboration meeting

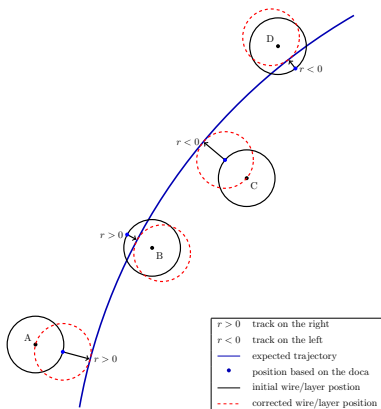
ALERT AHDC tracking
and alignment

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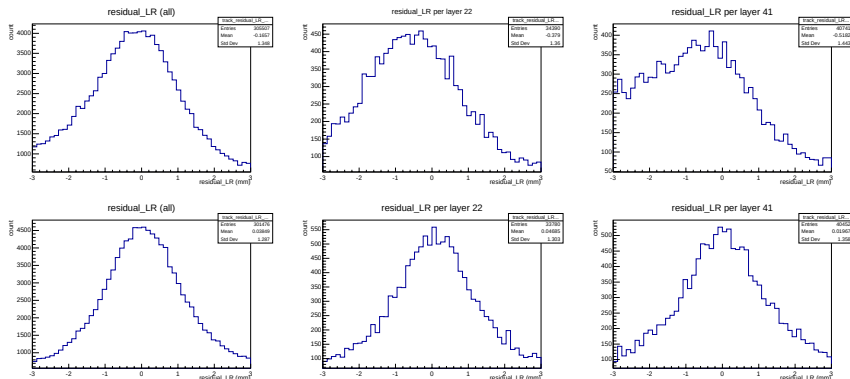
June 30, 2026

- ▶ Context: limitations in the AHDC track reconstruction
 - need to improve the overall tracking resolution
 - need to improve the resolution of the reconstructed θ
 - need to extra the relative position of the AHDC with respect to CLAS
- ▶ Principle:
 - Select elastic events : electron + track (collection of hits)
 - Compute the expected track kinematics from the electron ones
 - Propagate the expected track
 - Estimate how far it deviates from the measured collection of hits
- ▶ Our observable : **residual_LR**
 - It indicates whether the track is on right or on the left of the measurement
- ▶ Correction angles (rotZ)



$$\alpha_i = \alpha_{i-1} - \frac{r_i}{R} \cdot \frac{1}{2}$$

- First attempt (top → without alignment, bottom → with alignment)
 - The distributions are more centered
 - Move from -0.51 to 0.01 as mean value on layer 41
 - **residual_LR** is a good observable



Layer	11	21	22	31	32	41	42	51
rotZ (deg)	0	+2.55	+2.11	0	-0.32	+3	+2.13	-0.21

- ▶ The alignment is so much dependent on the precision of the vertex determination

$$(\text{AHDC vz}) = (\text{CLAS electron vz}) - (\text{vz shift})$$

- ▶ Limitations

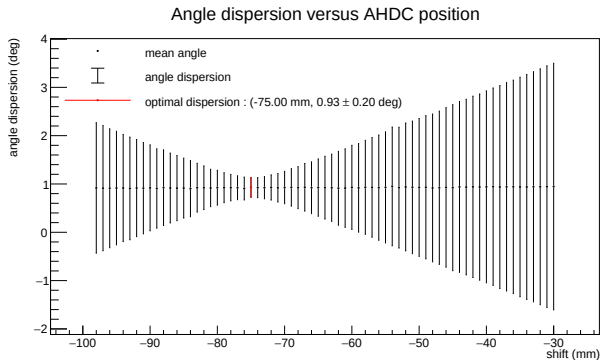
- (CLAS electron vz) has a small resolution for elastic events, $\theta_e \sim 6^\circ$
- (vz shift) relative position of the AHDC relative to the center of CLAS

- ▶ Solutions/Observations

- Doing a fit can release the constraint on (CLAS electron vz)
- Using the exact (vz shift) value make the correction angles smaller (so more physical)

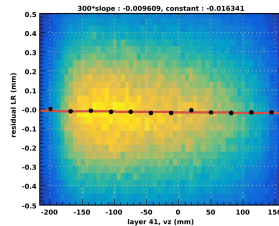
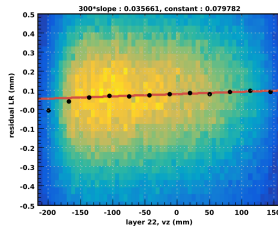
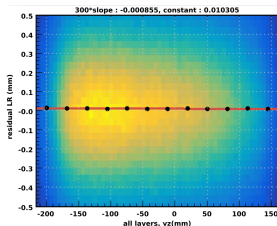
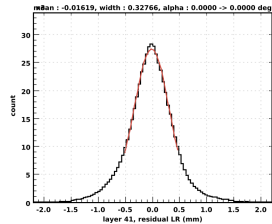
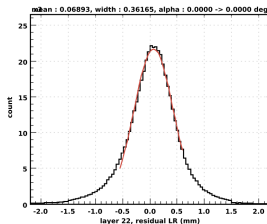
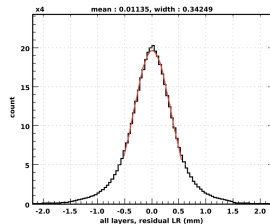
layer \ shift	-54 mm	-30 mm	-70 mm
11	0 deg	-1.49	+0.756
21	+2.55	+3.82	+1.517
22	+2.11	+3.24	+0.984
31	0	-1.54	+0.807
32	-0.32	-1.96	+0.382
41	+3.0	+3.73	+1.305
42	+2.13	+3.3	+0.975
51	-0.21	-1.61	+0.679

- According to the layer alignment without fit, the optimal AHDC position with respect to CLAS is **-75 mm**

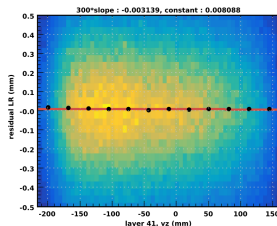
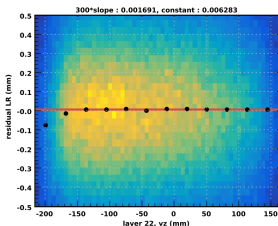
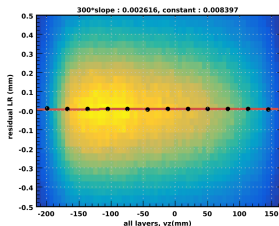
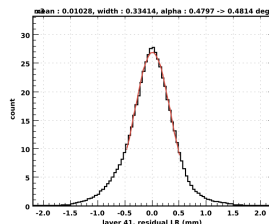
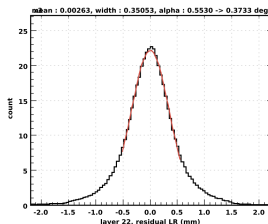
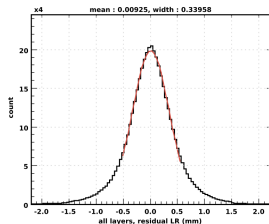


- The next alignment has been done by performing a **fit** with a **-75 mm** shift

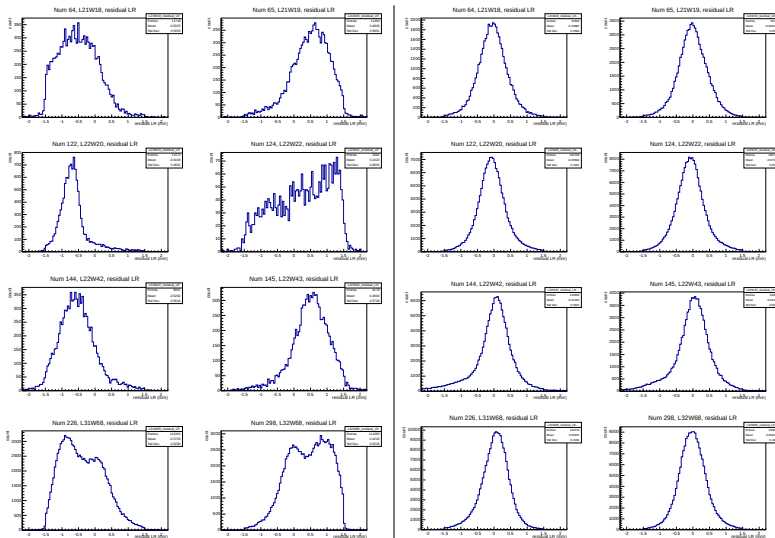
- ▶ The residual_LR shift is difficult to see in the integrated distribution
- ▶ A misalignment is highlighted looking at the residual_LR versus v_z
- ▶ We decided to apply different correction angles at the two ends of the detector



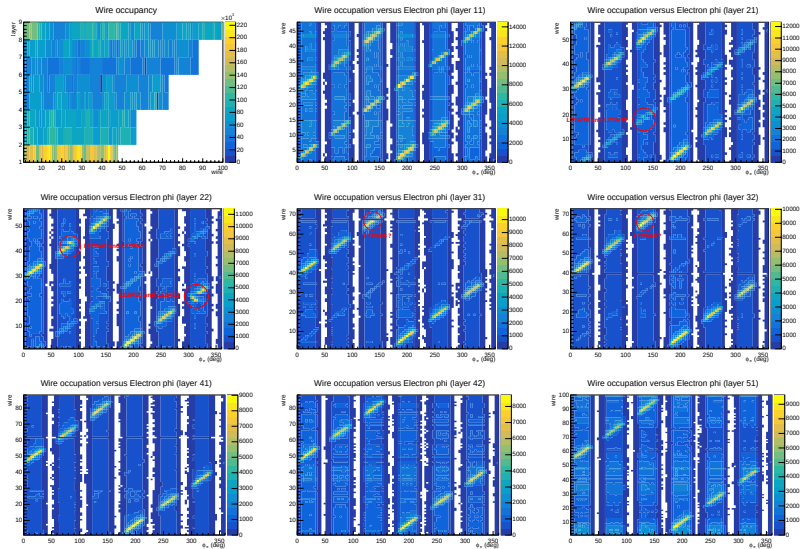
- ▶ Corresponding distributions after alignment
- ▶ Work already implemented in coatjava
- ▶ Next step: wire by wire alignment



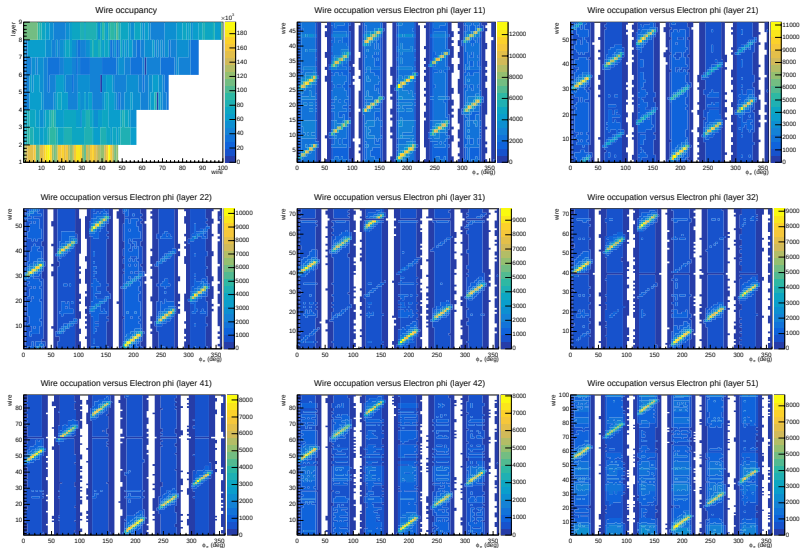
- Analyzing wire by wire distributions we found 4 couples of wire swaps
- Distributions before (on the left) and after fix (on the right)



- Swaps were also visible in the wire versus phi distributions for elastic events



► Wire versus phi distributions after fix



- t_0 and T2D calibration are ongoing, but we can already achieve good resolutions layer by layer, especially layer 41

